

Student Medical Centre

Optimizing healthcare and wellness through organized data structures

Data Structures & Algorithms 1 — Practical Assignment

Team — Group 7

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- D L S R Jayasekara

Student Registration

Doubly Linked List

- Inserting new students details
- Traversing list for the correct position
- Always sorted

Why Doubly Linked List?

Instant Insertion

Insertion anywhere in the sorted list without shifting other records

Search & Traverse

Bidirectional Traverse to locate a student by index number or name

HAV ABEYWARDENA

Patient Appointments



Queue

— **First-In, First-Out** —

Absolute fairness by arrival time.



Dual Queues

— **Normal & Urgent** —

Emergencies are prioritized without breaking the system

Calculating Average Time & Cost

Queue

Assigning temporary tokens to patients with Arrival Time

Arrival time, Discharge Time, Cost

Discharge Time and Cost is recorded for each patient

Avg Cost and Time/ Patient

Average cost for each patient and average time is calculated

MTNA PERERA

Lab Report Retrieval



Doctors review the most recent lab reports

Temporarily organizes retrieved reports for doctor's use



Stack — LIFO

Last-In, First-Out — Doctors see the **most recent** test results first via **POP**

DLSR JAYASEKARA

National Hospitals Within Proximity



Add / Update Hospital

Array for a static dataset



Search & Filter

Linear search by given
specialization



Sort by Distance

Insertion Sort to prioritize
closest hospitals

Frequent Medical Issues

Dynamic Storage

- **Singly Linked List** stores new issues as they appear
- Always ordered, **Traversed** from highest danger

Medical Issues Ranked

- Ranked by **occurrence count** using **Selection Sort**
- **Not stable**, not required

KLGV RANDINU

Pharmaceutical Suppliers



Organize the Suppliers

Can **add/remove** and **search** the suppliers



Singly Linked List

Insertion anywhere without shifting other records



Sort by Cost

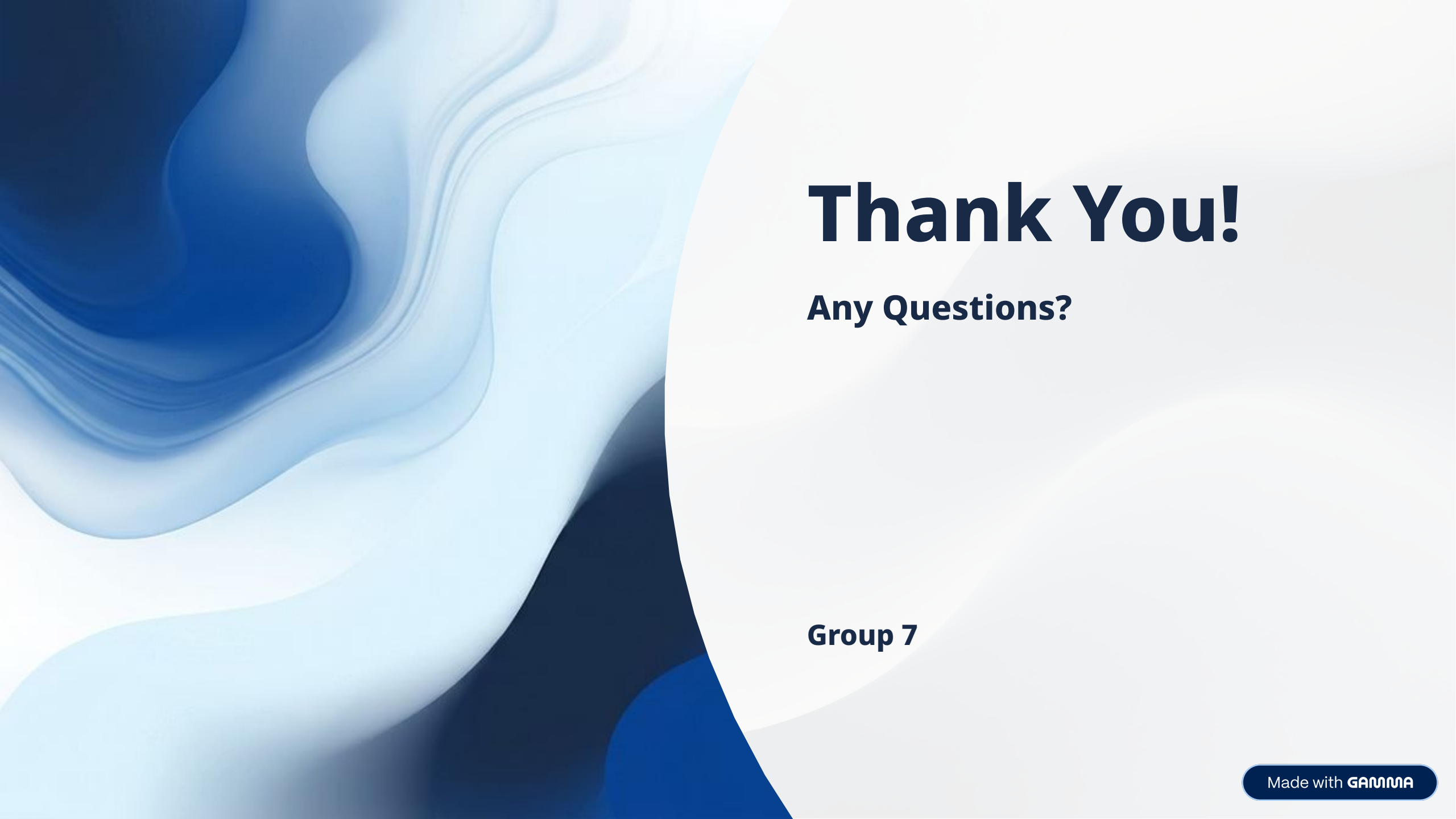
Sorts suppliers by **price** & **distance** as tiebreaker using bubble sort

Conclusion: Data Structures and Functions

Most Common Ailments
Average Patient Time, Cost
**Registration, Appointment
Number**
Lab Reports
Nearby Hospitals, Suppliers



Queue
Singly Linked List
Stack
Doubly Linked List
Array



Thank You!

Any Questions?

Group 7